Magnetosphere and changing fields

antocciphysics | TEACHER KEY | CPA

Learning target: Use evidence to distinguish a magnetic-field model from an unsupported space-weather claim.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 13; 3 course days later.*
New field is 1/3 of the original under the long-wire model.

02. Explain a worked example (worked)
Change field magnitude/orientation or loop area; relevant quantity is flux change through the loop.

03. Look and explain (model)
Changing flux through coil; emf around loop.

04. Build the solution (practice)
4 V in the proportional model.

05. Use the idea (reason)
A credible physical mechanism, measured/predicted field change and timing, source attribution, conductor geometry and current estimates; proximity alone is insufficient.

06. Choose a method (mixed) *Review: Unit 7, Day 4; 12 course days later.*
6 J of electric potential-energy difference per coulomb.

07. Exit check (exit)
No; a flux change is required.

08. STRETCH: Apply it another way (stretch)
Larger area gives greater emf for the same uniform dB/dt and one turn; real geometry and conductivity matter.

Teaching note: Science reference: NASA, “Solar Storms and Flares” and “Living with a Star: NASA and Partners Survey Space Weather Science.” Asteroid scenarios here are fictional.

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01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 13; 3 course days later.*
$B_f/B_i = 3/5 \approx 0.6$, using $B \propto I/r$.

02. Explain a worked example (worked)
Change field magnitude/orientation or loop area; relevant quantity is flux change through the loop.

03. Represent and interpret (model)
Changing flux through coil; emf around loop.

04. Solve and justify (practice)
2.5×10^{-5} V; no, it only follows from the supplied hypothetical change.

05. Analyze the reasoning (reason)
A credible physical mechanism, measured/predicted field change and timing, source attribution, conductor geometry and current estimates; proximity alone is insufficient.

06. Choose a method (mixed) *Review: Unit 7, Day 4; 12 course days later.*
32 J and 48 J; total 80 J.

07. Exit check (exit)
Real networks have geometry, grounding, resistance, controls and spatially varying fields; a justified larger model and observations are required.

08. STRETCH: Apply it another way (stretch)
$\text{Emf} = A\Delta B/\Delta t = 0.0004$ V. Real grids involve geometry, ground conductivity, spatial/temporal variation and network response.

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